XRPL Validator — Daily Engineering Report · 2026-08-28

Period: 2026-08-28 00:00 to 24:00 US Eastern (2026-08-28 04:00 to 2026-08-29 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 2 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
03:05 UTC	xrpl-validator	`3f9c141`	feat(xrpl-node): validations carry ServerVersion + Cookie; backport kVfFullValidation to canonical tree
03:16 UTC	xrpl-validator	`3f5a198`	feat(xrpl-node): report our own version — HV 1.0.0, not rippled's

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-node): probe s2 fallback retries transient failures** (`24c2d93` , xrpl-validator) Single-attempt fallback silently kept the primary's lgrNotFound whenever s2 dropped the call (fleet contention: 2428 permanently-uncacheable misses on l106281182, phantom 52/57). Retry 3x like the primary, treat slowDown-class JSON errors as transient, settle on success or a deterministic negative. lgrNotFound from the full-history server itself is settled. DX_RPCLOG gains a FALLBACK-> outcome receipt.
- **feat(xrpl-ledger): BookStep requireAuth — unauth owners reaped, unauth pools yield no synthetic** (`46f139b` , xrpl-validator) rippled runs requireAuth(book.in, owner) per offer in the stream (BookStep.cpp:755): under the issuer's lsfRequireAuth the OWNER — an AMM pool account included — must hold the ISSUER-side auth flag (issuer low ? lsfLowAuth : lsfHighAuth, the 0x40000/0x80000 line bits) on its line for the asset flowing INTO it; a missing line is tecNO_LINE, and there is NO balance grandfather (that mercy is only DirectStep.cpp:430's …
- **fix(xrpl-node): probe fallback triggers on 'internal' — rippled masks lgrNotFound for non-admin** (`dffa32c` , xrpl-validator) the reference node answers the same account_info@pre-window ledger 'lgrNotFound' to an admin curl and 'internal' to the validator host; public s2 answers 'internal' where Clio-backed xrplcluster serves the data. The lgrNotFound-only trigger meant the fallback NEVER fired from the gates — the entire 18-fixture 'deterministic' hist residue was hydration that never got a chance (ours tecNO_DST vs net tecNO_PERMISSION with the dst account provably …
- **fix(xrpl-ledger): NFT transfer-fee royalty rounds half-even, not floor** (`84b245c` , xrpl-validator) rippled computes the royalty as multiply(amount, transferFeeAsRate(fee)) (NFTokenAcceptOffer.cpp:420/:540); asAmount(rate) is non-native so the native fast path never applies and the Number pipeline runs: ONE half-even

reduction of the FULL product to 16 significant digits (multiply folds the whole remainder into the tie), /100000 as pure exponent arithmetic, then half-even again at the integer-drop boundary. We …

- **feat(xrpl-node): validations carry ServerVersion + Cookie; backport kVfFullValidation to canonical tree** (`3f9c141`, xrpl-validator) sfServerVersion (UInt64 field 11) encoded per BuildInfo:: encodeSoftwareVersion — 0x183B identifier, 3.3.0, release bit — fills the empty Version column on VHS/xrpscan/livenet. sfCookie (UInt64 field 10): one random per process run, rippled's restart disambiguator. Both sit canonically between SigningTime and LedgerHash.
- **feat(xrpl-node): report our own version — HV 1.0.0, not rippled's** (`3f5a198`, xrpl-validator) The ServerVersion identifier becomes 0x4856 (ASCII 'HV', Halcyon Validator) with version 1.0.0: this is an independent Rust engine tracking rippled 3.3.0's protocol surface, and claiming to BE rippled 3.3.0 misrepresented what runs here. A foreign identifier is inert on-network (BuildInfo::isNewerVersion tests isXrpldVersion first), and VHS appears to decode semver bits regardless of identifier (44 mainnet …

Independence program

No independence record for UTC day 2026-08-28; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Sponsor-Capped Batch for Game Economies** (gaming) Add a `tfSponsoredBatch` flag plus `sfSponsorID` / `sfSponsorCap` SFields to the outer Batch (XLS-56) transaction; when set, aggregate fees and any

owne

- **Segregated Sub-Accounts for Atomic DvP** (settlement) Add a new owner-reserved ledger object `ltSUBACCOUNT` (keylet::subAccount(parent, subID)) that holds its own MPToken/RippleState balances and authoriz
- **Unified Content-Addressed Token Metadata** (tokens) Add a reserve-free, content-addressed ledger object `ltMETADATA` (keylet::metadata(sha256(content))) created by a new `MetadataSet` transaction carryi

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-28 17:26	—	xrpl-validator	`24c2d93`	fix(xrpl-node): probe s2 fallback retries transient failures
08-28 19:32	—	xrpl-validator	`46f139b`	feat(xrpl-ledger): BookStep requireAuth — unauth owners reaped, unauth pools yield no synthetic
08-28 22:15	—	xrpl-validator	`dffa32c`	fix(xrpl-node): probe fallback triggers on 'internal' — rippled masks lgrNotFound for non-admin
08-28 22:41	—	xrpl-validator	`84b245c`	fix(xrpl-ledger): NFT transfer-fee royalty rounds half-even, not floor
08-29 03:05	08-29 03:05	xrpl-validator	`3f9c141`	feat(xrpl-node): validations carry ServerVersion + Cookie; backport kVfFullValidation to canonical tree
08-29 03:16	08-29 03:16	xrpl-validator	`3f5a198`	feat(xrpl-node): report our own version — HV 1.0.0, not rippled's

Times in UTC. A commit appears on the day it was committed to main or merged into it.